

Find worlds worth looking at.

A calmer way into the archive: start with recognizable planet stories, then open the lab only when you want the measurements behind them.

Browse worlds

See science lenses

Open lab

The exact archive values are still here; they are just tucked behind a more readable first layer.

FEATURED ROUTE

WORLD PROFILE

TEMPERATE TRAIL

ATMOSPHERE PICKS

STRANGE WORLDS

habitable-zone candidates

transit-friendly worlds

nearby systems

DATASET STATUS

NASA archive nasa-tap catalog from NASA Exoplanet Archive TAP service is loaded. Exact values live in the science lab.

[Open JSON data](#)

SCIENCE LENSES

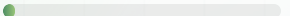
Choose a lens before you open the lab

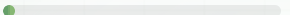
Start with the kind of world you want to understand. The archive still has the precise measurements, but this layer reads like a guidebook instead of a spreadsheet.

Temperate routes

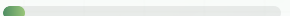
climate paths

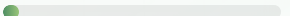
Find worlds that sit near the archive's habitable-zone story.

CONSERVATIVE HZ  RARE

OPTIMISTIC HZ  RARE

TOO IRRADIATED  DOMINANT

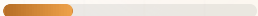
TOO LITTLE FLUX  RARE

UNKNOWN  RARE

World families

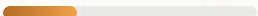
visual family

Move between rocky worlds, sub-Neptunes, giants, and uncertain records.

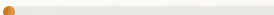
TERRESTRIAL  VISIBLE

SUPER-EARTH  COMMON

SUB-NEPTUNE  DOMINANT

GIANT  VISIBLE

SUPER-JUPITER  DOMINANT

UNKNOWN  RARE

Detection stories

observing paths

See whether a world revealed itself through shadow, wobble, lensing, or direct light.

TRANSIT

DOMINANT

RADIAL VELOCITY

VISIBLE

MICROLENSING

RARE

IMAGING

RARE

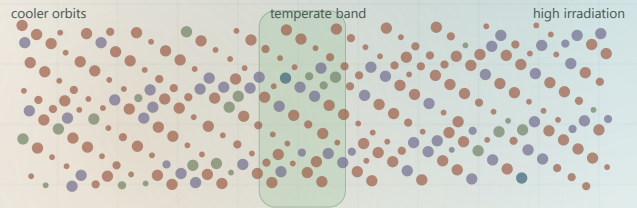
TRANSIT TIMING
VARIATIONS

RARE

Survey terrain

target landscape

A map-like view of where worlds fall from cool prospects to high-irradiation outliers.



Archive eras

discovery waves

Follow the catalog from early ground-based records into modern survey finds.

EARLY SEARCHES

RARE

SURVEY BOOM

DOMINANT

CURRENT ERA

COMMON

ARCHIVE UNKNOWNNS

RARE

WORLD BROWSER

Browse planets by story

Start from featured worlds, filter by science case, and open any card for the measured readout.

Reset filters

Search planet or host

TRAPPIST, Kepler, Proxima...

Discovery method

All methods

Size class

All sizes

Science filter

All science cases

Discovery era

All years

Sort by

Featured worlds

Matching worlds

Click a card to analyze it in the lab.



Proxima Cen b

Around Proxima Cen

Conservative HZ

A compact world in the temperate conversation.

Climate

inside main
temperate band

Body type

rocky-
compatible

Signal

clear transit

Follow-up

measurable
wobble



Barnard b

Around Barnard's star

Terrestrial

A catalog world ready for closer inspection.

Climate

highly
irradiated

Body type

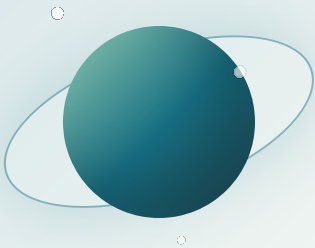
volatile-rich
candidate

Signal

clear transit

Follow-up

delicate wobble



Teegarden's Star b

Around Teegarden's Star

Optimistic HZ

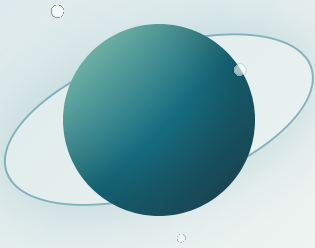
A compact world in the temperate conversation.

Climate
**near the
temperate edge**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



Teegarden's Star c

Around Teegarden's Star

Conservative HZ

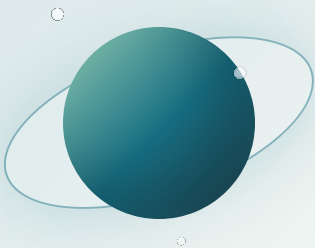
A compact world in the temperate conversation.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



TRAPPIST-1 e

Around TRAPPIST-1

Conservative HZ

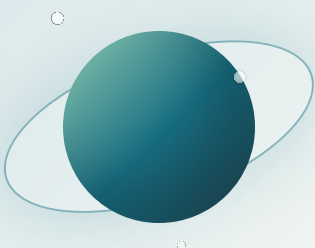
A compact world in the temperate conversation.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



TRAPPIST-1 f

Around TRAPPIST-1

Conservative HZ

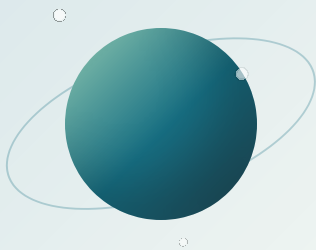
A compact world in the temperate conversation.

Climate
**inside main
temperate band**

Body type
**rocky-
compatible**

Signal
**very strong
transit**

Follow-up
**measurable
wobble**



Kepler-186 f

Around Kepler-186

Conservative HZ

A compact world in the temperate conversation.

Climate

**inside main
temperate band**

Body type

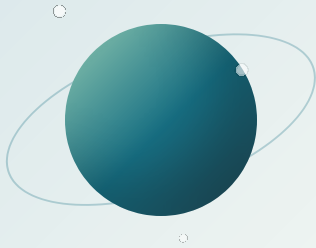
**rocky-
compatible**

Signal

subtle transit

Follow-up

delicate wobble



TOI-700 d

Around TOI-700

Conservative HZ

A compact world in the temperate conversation.

Climate

**inside main
temperate band**

Body type

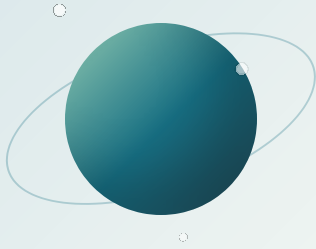
**rocky-
compatible**

Signal

clear transit

Follow-up

delicate wobble



K2-18 b

Around K2-18

Optimistic HZ

A catalog world ready for closer inspection.

Climate

**near the
temperate edge**

Body type

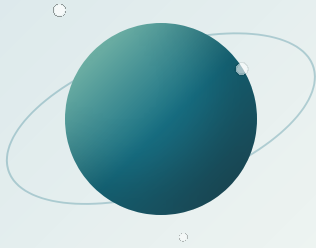
**volatile-rich
candidate**

Signal

clear transit

Follow-up

**measurable
wobble**



LHS 1140 b

Around LHS 1140

Conservative HZ

A dense world with useful mass-radius context.

Climate

**inside main
temperate band**

Body type

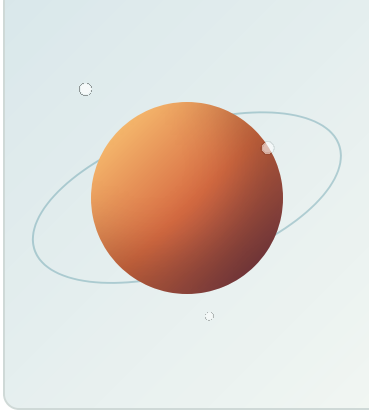
**rocky-
compatible**

Signal

**very strong
transit**

Follow-up

**measurable
wobble**



GJ 486 b

Around GJ 486

Super-Earth

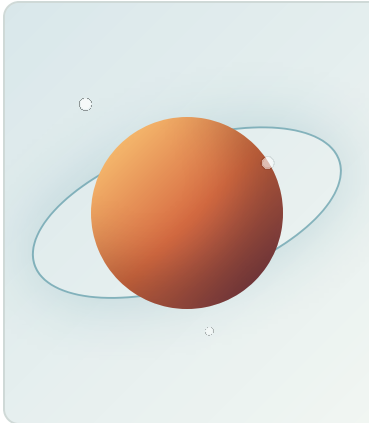
A dense world with useful mass-radius context.

Climate
highly irradiated

Body type
rocky-compatible

Signal
clear transit

Follow-up
measurable wobble



GJ 1214 b

Around GJ 1214

Sub-Neptune

A strong candidate for atmosphere-focused follow-up.

Climate
highly irradiated

Body type
volatile-rich candidate

Signal
very strong transit

Follow-up
measurable wobble

Load more

SELECTED PLANET

Proxima Cen b

Proxima Cen · Terrestrial · Radial Velocity

Method

Radial Velocity

Discovery era

space-survey wave

Size family

Terrestrial

System distance

nearby system

Host star

cool red host

Facility

European Southern Observatory

Science readout

Climate read

inside main temperate band

Bulk character

rocky-compatible

Transit signal	clear transit
RV follow-up	measurable wobble
Atmosphere signal	possible spectrum
Composition	Rocky-compatible bulk density

No curated description is available for this record. Refreshing from NASA TAP preserves core physical and discovery fields.

Analyze in science lab

OBSERVATORY NOTEBOOK

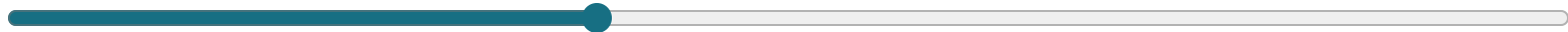
Selected-world observing notes

The science figures are generated by Python/Matplotlib during the data-refresh workflow, then served as responsive SVGs that use the full browser width where possible. Change albedo and atmospheric mean molecular weight to update the numerical readout; the plotted observing diagnostics use catalog parameters.

Planet

Proxima Cen b — Proxima Cen

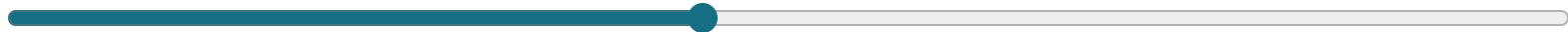
Bond albedo: 0.30



Atmosphere model

Earth-like N₂/O₂, $\mu \approx 29$

Transmission scale heights: 5



Proxima Cen b

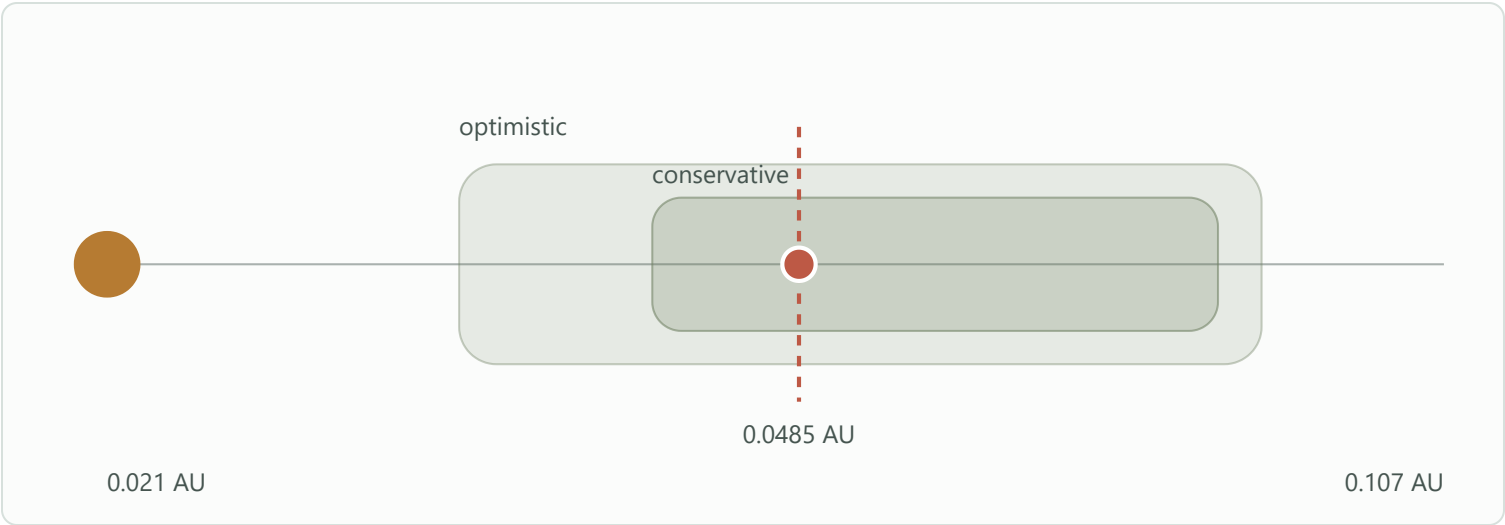
Proxima Cen · Radial Velocity · 2016

Inside conservative HZ

Incident flux 0.641 S$\oplus$ Luminosity divided by orbit squared	Model T _{eq} 227.9 K albedo 0.30	Bulk density 5.46 g/cm³ Rocky-compatible bulk density	Surface gravity 1.01 g$\oplus$ vesc 11.38 km/s
Transit depth 4,398 ppm central duration $\approx$ 1.23 h	RV semi-amplitude 1.226 m/s edge-on unless inclination is known	Transmission proxy 44.4 ppm H $\approx$ 6.6 km	Astrometric wobble 0.967 μas maximum angular reflex signal
Stellar luminosity 0.002 L$\odot$ R $\star$ 0.141 R $\odot$	Semi-major axis 0.0485 AU P 11.18 d	Rocky CMF proxy 30.8% valid mainly for 1–8 M $\oplus$ rocky planets	Archive T _{eq} 218 K literature/archive field

Habitable-zone placement

orbit vs. climate boundaries



Transit light curve

limb-darkened geometry

Normalized stellar flux



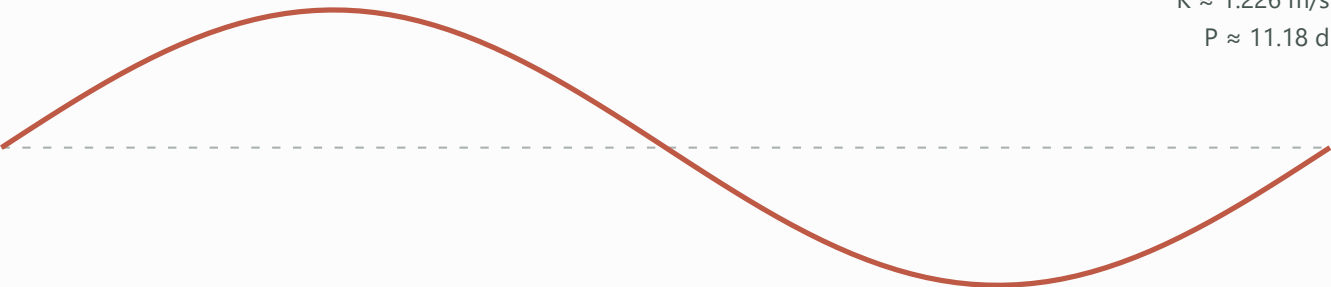
depth $\approx 4,398$ ppm
duration ≈ 1.23 h

Orbital phase around transit

Radial velocity curve

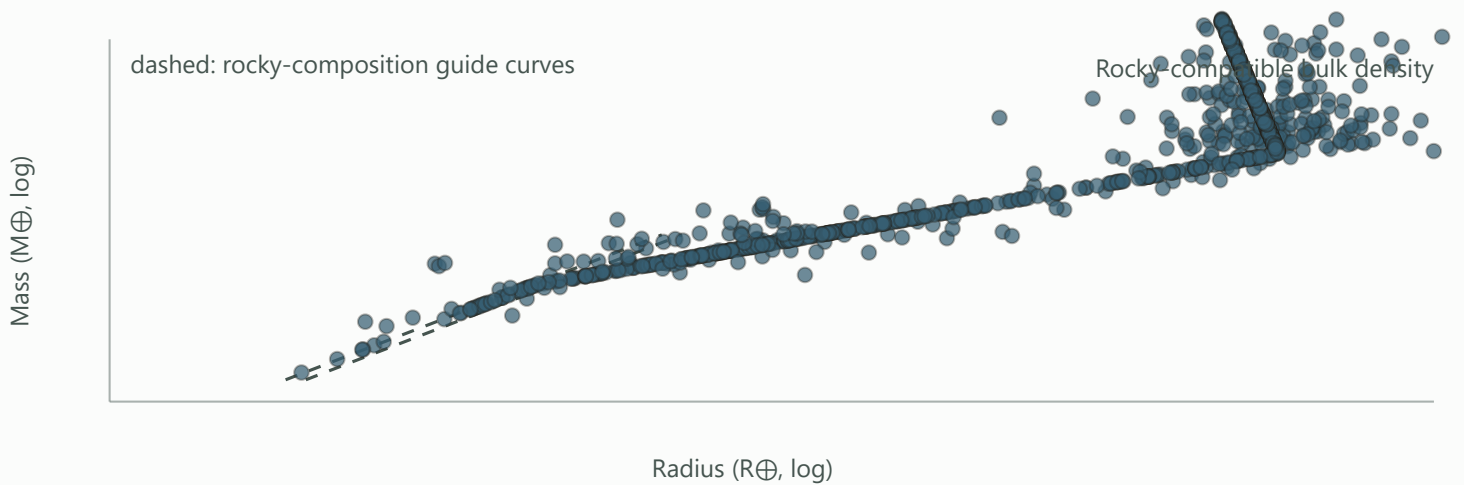
Keplerian model

Radial velocity relative to system barycenter



$K \approx 1.226$ m/s
 $P \approx 11.18$ d

Orbital phase



Assumptions and caveats

- Transit duration assumes a circular, central transit and ignores limb darkening.
- RV amplitude uses a simplified Keplerian semi-amplitude and assumes $\sin(i)=1$ when inclination is unknown.
- Transmission signal is a scale-height proxy, not an atmospheric retrieval or detectability guarantee.
- The lab figures prefer static Python/Matplotlib SVGs generated during data refresh; browser SVG fallbacks are used only when a target-specific plot pack is unavailable.

EXPLAINABLE ASTRONOMY

A compact exoplanet physics chapter

The interface now treats the model layer like a textbook page: display equations, symbols, units, and assumptions are separated from the prose. These are simplified observing equations for public science and target triage, not a replacement for a full retrieval or orbital fit.

- ★ stellar quantity
- p planetary quantity
- ⊕ Earth-normalized unit
- ☉ solar-normalized unit

01

Incident stellar flux

How strongly the planet is irradiated relative to Earth.

DISPLAY MATH

$$S_p = \frac{L_\star/L_\odot}{(a/\text{AU})^2} \quad (1)$$

A planet with $S_p = 1$ receives the same bolometric flux Earth receives from the Sun. This is the quantity used for the radius-flux population map and for a first-order climate triage.

 $L_\star$

stellar luminosity

 a

semi-major axis

 S_p

planet incident flux in Earth units

02

Equilibrium temperature

A zero-dimensional temperature estimate controlled by albedo.

DISPLAY MATH

$$T_{\text{eq}} = T_{\text{eff}} \left(\frac{R_\star}{2a} \right)^{1/2} (1 - A_B)^{1/4} \quad (2)$$

The albedo slider changes A_B . The expression assumes global heat redistribution and no greenhouse correction, so it is a diagnostic temperature rather than a climate model.

 T_{eff}

stellar effective temperature

 $R_\star$

stellar radius

 A_B

Bond albedo

03

Kopparapu-style habitable-zone boundaries

Climate boundaries expressed as stellar-flux limits and then converted to orbital distance.

$$S_{\text{eff}} = S_{\text{eff},\odot} + aT_{\star}' + bT_{\star}'^2 + cT_{\star}'^3 + dT_{\star}'^4 \quad \text{DISPLAY MATH} \quad (3a)$$

$$d_{\text{HZ}} = \left(\frac{L_{\star}/L_{\odot}}{S_{\text{eff}}} \right)^{1/2} \text{ AU}, \quad T_{\star}' = T_{\text{eff}} - 5780 \text{ K} \quad \text{DISPLAY MATH} \quad (3b)$$

The site uses inner and outer flux limits to classify a planet as inside, hotward of, or coldward of the optimistic and conservative habitable-zone bands.

The plotted HZ panel shows the orbit against those boundaries; it does not imply habitability without atmosphere, geophysics, stellar activity, and composition constraints.

04

Transit depth and duration

Photometric observability from the ratio of planetary and stellar radii.

$$\delta_0 \simeq k^2, \quad k = \frac{R_p}{R_{\star}} \quad \text{DISPLAY MATH} \quad (4a)$$

$$T_{14} \simeq \frac{P}{\pi} \arcsin \left[\frac{R_{\star}}{a} \frac{\sqrt{(1+k)^2 - b^2}}{\sin i} \right] \quad \text{DISPLAY MATH} \quad (4b)$$

The Python light-curve figure adds a simple limb-darkened shape around this geometric depth. The catalog readout reports depth in parts per million.

k

radius ratio

b

impact parameter

i

orbital inclination

05

Radial-velocity semi-amplitude

The stellar reflex velocity induced by an orbiting planet.

DISPLAY MATH

$$K = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_p \sin i}{(M_\star + M_p)^{2/3}} \frac{1}{\sqrt{1 - e^2}} \quad (5)$$

When inclination is missing, the app uses an edge-on approximation. The plotted curve is a Keplerian observing model from catalog parameters, not a fit to raw spectra.

Earth-unit shortcut: $K \approx 0.08945 \text{ m s}^{-1} (M_p/M_\oplus)(P/\text{yr})^{-1/3} (M_\star/M_\odot)^{-2/3} \sin i / \sqrt{1 - e^2}$.

06

Bulk density, gravity, and escape velocity

First-order structural diagnostics from mass and radius.

DISPLAY MATH

$$\rho_p = \rho_\oplus \frac{M_p/M_\oplus}{(R_p/R_\oplus)^3}, \quad g_p = g_\oplus \frac{M_p/M_\oplus}{(R_p/R_\oplus)^2} \quad (6a)$$

DISPLAY MATH

$$v_{\text{esc},p} = v_{\text{esc},\oplus} \left(\frac{M_p/M_\oplus}{R_p/R_\oplus} \right)^{1/2} \quad (6b)$$

These quantities help separate likely rocky worlds, volatile-rich sub-Neptunes, and gas giants. They are most meaningful when mass and radius are both measured rather than inferred.

Transmission-spectroscopy signal proxy

A scale-height estimate for atmosphere follow-up prioritization.

$$H = \frac{k_B T_{\text{eq}}}{\mu m_H g_p} \quad (7a)$$

DISPLAY MATH

$$\Delta_\lambda \sim \frac{2NH R_p}{R_\star^2} \quad (7b)$$

DISPLAY MATH

The atmosphere menu changes μ , the mean molecular weight. The scale-height slider changes N , the number of scale heights assumed to contribute to spectral modulation.

This is intentionally a proxy: clouds, hazes, composition, stellar contamination, and instrument systematics can dominate real observations.

HOW EXOPLANETS ARE DETECTED

Discovery methods connected to observables

Each method measures a different physical signature, so the science lab reports the quantities that matter for follow-up rather than only the discovery label.

Transit photometry

Detects a small, periodic dimming when a planet crosses the stellar disk. The depth constrains radius; repeated events constrain period; duration and shape constrain orbital geometry.

Radial velocity

Measures Doppler shifts in stellar spectra caused by the star orbiting the system barycenter. It gives a mass-related signal and is strongest for massive planets close to low-mass stars.

Direct imaging

Separates faint planetary light from a bright host star. It favors young, hot, massive planets at wide separations where contrast and angular separation are less punishing.

Microlensing

Uses the gravitational field of a foreground system as a lens. It can reveal planets far away, but the events are usually non-repeatable and follow-up characterization is difficult.

OPEN DATA WORKFLOW

Built for reproducible public publishing

The repository includes a Python script that queries the NASA Exoplanet Archive TAP service, converts selected fields into a browser-ready JSON catalog, and keeps the static site deployable through GitHub Pages.

[View repository](#)

1. Static site

No build step is required. The browser loads `index.html`, `styles.css`, `app.js`, and `data/exoplanets.json`.

2. Science fields

The updater requests stellar mass, stellar radius, luminosity, inclination, eccentricity, insolation, density, and photometric fields when available.

3. Data refresh

Run `python scripts/fetch_exoplanets.py` locally or trigger the included GitHub Actions workflow to update and redeploy the JSON catalog.

[NASA Exoplanet Archive](#)[NASA TAP documentation](#)[Kopparapu HZ calculator](#)[NASA Exoplanets](#)

Built by Arif Solmaz as an open, public GitHub Pages exoplanet observatory notebook. Use the included archive workflow for official NASA-derived values; treat browser-side estimates as educational approximations.